

COURSE INFORMATION

School/Faculty:	Computing/Engineering	Page:	1 of 7
Program name:	Bachelor of Computer Science (Software Engineering)		
Course code:	SECP3204	Academic Session/Semester:	2024/2025/1
Course name:	Software Engineering	Pre/co requisite (course name and code, if applicable):	-
Credit hours:	4		

Course synopsis	This course is designed to give students an introduction to an engineering approach in the development of high-quality software systems. It will discuss the important software engineering concepts in the various types of the common software process models. The students will also learn the concepts and techniques used in each software development phase including requirements engineering, software design and software testing. This course will also expose the students to utilizing object-oriented method (e.g. UML) and tools in analyzing and designing the software. At the end of this course, students are expected to be able to appreciate most of the common software engineering concepts and techniques as well as producing various software artifacts and deliverables.		
Course coordinator (if applicable)	Dr Aryati Bakri		
Course lecturer(s)	Name (Section No.)	Section	E-mail
	Dr Aryati Bakri	01	aryati
	Dr Muhammad Iqbal Tariq Idris N28a - miqbaltariq	02	miqbaltariq

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (Code)	Taxonomies & generic skills	T&L methods	Assessment methods
CLO1	Analyze the appropriate object-oriented method and software process models to solve software engineering case study.	PLO1 (KW)	C4	KWL chart, Jigsaw-activity, Mind-mapping, In-class problem solving	Quiz Test Final Exam
CLO2	Propose a suitable solution of software architecture, object-oriented design model and test cases using state-of-the-art methods and tools for a software engineering case studies.	PLO3 (PS)	C6, P3, P4	Group Discussion Think-Pair-Share, Project - case study	Assignment 1 Assignment 2 Assignment 3 Assignment 4

Edited by:	Certified by:
Name: Noraini Ibrahim	Name:
Signature:	Signature:
Date: 05 October 2023	Date:

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Course code:	SCSP 2304	Academic Session/Semester:	2024/2025/1
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CLO3	Ability to do work effectively in a team and comprehend the interchangeable role of leaders and followers with team members.	PLO7 (TW1, TW2)	A5	Collaborative/team discussion, Peers teamworking reflection using myPeer app, Self-reflection on leadership (video-LinkedIn)	Peer-Team Working Assignment 1 Assignment 2 Assignment 3 Assignment 4 Industry
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Details on Innovative T&L practices:

No.	Type	Implementation
1.	KWL (Know-What-Learn) chart	Conducted activity to engage students in a new topic, activate prior knowledge, share unit objectives, and monitor students' learning.
2.	Jigsaw-activity	Conducted collaborative group activity where students effectively teach each other (with the teacher's guidance) a given skill or procedure, topic or problem.
3.	Mind-mapping	Conducted activity to visually organize information.
4.	Project - case study	Conducted based on given real world problem. The students have to manage their projects by submitting specified deliverables based on the given problem. The problems are given in sequential steps throughout the semester.
5.	Collaborative Group Discussion	Students use an online platform to interact with peers by posting thoughts or comments in response to their project discussion.
6.	Peers teamworking and Self leadership reflection	Students assessing their own work and provide feedback to each other on their teamwork and leadership quality
7.	Work Based Learning (WBL)	Conducted based on given real world problem. The students must manage their projects by submitting specified deliverables based on the given problem.

Weekly Schedule

W1 6 - 10 Oct	Module 1: Introduction to Software Engineering Software definition, software engineering as a layered technology, types of software, inherent difficulties in software engineering, SE quality focus, Documentation Standard
W2 13 - 17 Oct	Module 2: Software Process Model Plan Driven or Agile Process Model, general software process model waterfall, incremental, reuse oriented, software process model which copes with change spiral model and Rational Unified Process

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W3 20 - 24 Oct	Module 3: Agile Software Development Agile methods, Agile development techniques, Agile project management, scaling Agile methods Software Engineering project – Problem Formulation A1 Briefing Identify project context and content, identify process model (Plan-driven/Agile, planning the schedule of the project)
W4 27 - 31 Oct 31 Oct - Deepavalli	Module 4: Requirements Engineering Types of requirements, functional and non-functional requirements, requirements specification, requirements engineering processes
W5 3 - 7 Nov	Module 5: Requirements Analysis and modeling A1 Submission, A2 Briefing Use case modeling and specification, domain modeling activity diagram, sequence diagram, state chart/state machine diagram
W6 10 - 14 Nov	Software Engineering project – Requirements analysis and models Use case modeling and specification, domain modeling activity diagram, sequence diagram, state chart/state machine diagram (cont)
W7 17 - 21 Nov	Software Engineering project – Requirements analysis and models A2 Submission Produce requirements document that includes all the required analysis models
24 - 30 Nov	MID-SEMESTER BREAK (W8)
W9 1 - 5 Dec	Industry Day I/ Initial Design (Figma) Module 6: Architectural Design MIDTERM TEST, A3 Briefing Architectural design and detailed design, Architectural design decision, architectural view, the component diagram in relation to architectural view, architectural pattern: Model-View-Controller (MVC), layered, repository, client server and pipe filter
W10 8 - 12 Dec	Module 7: Object-Oriented Detailed Design Relationships among analysis, design and implementation, object-oriented design using UML, object-oriented design principles, elaborating UML diagrams from analysis

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W11 15 - 19 Dec	Software Engineering project – Design model Produce design document* that includes all the required design models
W12 – W13 22 - 26 Dec 25 Oct - Christmas 29 Dec 2024 - 2 Jan 2025	Module 8: Software verification, validation and testing A4 Briefing, A3 Submission Introduction to verification and validation, verification and validation planning, software inspections. System testing, Component testing, Test case design, Test Case Design using Black-box and white box
W14 5 - 9 Jan 2025	Software Engineering project – Test case design Produce testing document* that includes all the required test cases
W15 12 - 16 Jan 25	Project Presentation/ Industry Day II
W16 19 - 25 Jan 25	REVISION WEEK (1 week)

*Note: System Documentation template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of the course.

Transferable Skills (skills learned in course of study which can be useful and utilised in other settings)

Define software engineering problem and develop models and documentation based on identified software process model that can be used in students' future career as a software engineer.

Effective Learning Time (ELT) details

Distribution of effective		Teaching and Learning Activities	
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Learning Time (SLT), Course Content Outline							TOTAL ELT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to Face	
CLO	L	T	P	O			
CLO 1 – 50%	45h	10h		10h		7h	72h
CLO 2 – 40%	9h		9h	25h	5h	7h	55h
CLO 3 – 10%			4h	40h		4h	48h
Total ELT	54h	10h	13h	40h	5h	18h	175h

L: Lecture, T: Tutorial, P: Practical, O: Others

Continuous Assessment	TOTAL ELT	
	Percentage	
Quiz	5	1
Assignment 1- 4	36	12
Industry	10	4
Academic Paper	10	1
Teamworking	4	2
Test	15	2
Final Exam	20	3
Total ELT	18h	140h

1	Quiz (5%)	PLO1 (KW)	5%	As in CLO1
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2	Assignment 1 (4%) Proposal & Planning	PLO3 (PS)	Documentation (1%) Presentation (3%)	As in CLO2
3	Assignment 2 (6%) SRS	PLO3 (PS)	Documentation (1%) Presentation (5%)	As in CLO2
4	Assignment 3 (14%) SDD	PLO3 (PS)	Documentation (3%) Presentation (11%)	As in CLO2
5	Assignment 4 (12%) SDT	PLO3 (PS)	Documentation (2%) Presentation (10%)	As in CLO2
6	Industry	PLO7 (TW2)	10%	As in CLO3
7	Academic writing & Presentation	PLO1 (KW)	10%	As in CLO1
8	Teamwork	PLO7 (TW1)	Peer Review – Teamwork for each Assignment 1-4 4%	As in CLO3
Summative Assessment			Percentage	Total SLT
1	Test	PLO1 (KW)	15	As in CLO1 (2h)
2	Final Exam	PLO1 (KW)	20	As in CLO1 1h 2h
Grand Total			100	200h

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

UML modelling tools – Enterprise Architect or any open-source tools.

Learning resources:

Main references

Sommerville, I. (2016). "Software Engineering", 10th Edition, Pearson.

Additional references

Agarwal, U. (2012), "Software Engineering", 1st Edition, Indian: S. Kataria and Sons.

Tsui, F. and Karam, O., (2011), "Concise Guide to Software Engineering: From Fundamentals to Application Methods", USA: Springer.

Bruegge, B. and Dutoit, A. H. (2000), "Object-oriented Software Engineering Conquering Complex and Changing Systems", USA: Prentice Hall Publication.

Bennett, S., McRobb, S. and Farmer, R. (2010), "Object-oriented systems analysis and design using UML", 4th Edition, USA: McGraw-Hill.

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Arlow, J. and Neustadt, I. (2002), "UML and the Unified Process: Practical Object-Oriented Analysis & Design", UK: Pearson Education.

Selamat, A., Selamat, M. H. and Zainuddin, N. (2021), "Software Engineering", Malaysia: UTM Press.

Learning Management System (LMS)

<http://elearning.utm.my>

Academic honesty and plagiarism:

Project assignments are team-based activities. Copying of work (texts, simulation results etc.) from other teams or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

Disclaimer:

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